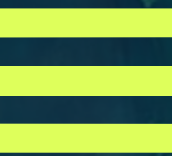


VOICE OF SECOPS 6TH EDITION 2025

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Cybersecurity & AI:

Promises, Pitfalls—and Prevention Paradise

RESEARCH REPORT

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Introduction

AI has become a routine part of doing business in 2025. It's changing how we operate across every functional area, including SecOps. **Almost three-quarters (72%) of organizations say they've revised their cybersecurity strategy over the past year because of AI, and 86% have increased their use of AI within security operations.**

The benefits of AI for cybersecurity are evident. **Organizations are saving, on average, 12 hours per week in manual processing time thanks to AI. More than half of organizations (58%) have rolled out AI-powered threat detection and response tools** to help protect their data.

However, when we delve deeper into the adoption, understanding, and operational response to AI, we uncover a more complex picture. Despite the transformational impact of the technology, **a quarter of organizations say they haven't changed their cybersecurity strategy at all from last year. Survey respondents ranked zero-day attacks as the least concerning risk**, despite the devastation they cause. And while AI is a boon for cybersecurity teams, **69% say the technology is also contributing to SecOps burnout.**

The vast range of technological capabilities encompassed by the term 'AI' is contributing to this complexity. Of the 500 senior cybersecurity experts we surveyed across the U.S., **fewer than a third (28%) understand the relationship between machine learning and deep learning.**

There is a strong correlation between this knowledge gap in certain sectors and exposure to cyber risk: for instance, **50% of critical infrastructure organizations suffered an AI-powered attack in the last year – the same sector that was least likely to be able to accurately distinguish between machine learning and deep learning.**

Different types of AI are used to address different issues: deep learning is a branch of machine learning, and one of the only ways to prevent known and unknown cyber threats in the face of cybercriminals equipped with AI-powered tools.

Protecting organizational data against increasingly complex and unpredictable threats will require organizations to truly understand and capitalize on this emerging technology in order to win the race against bad actors who are determined to do the same.

Methodology

Sapio Research surveyed 500 senior cybersecurity experts from companies with 1000+ employees in the U.S. The interviews were conducted online in April 2025 using an email invitation and an online survey.

Respondents worked at organizations which operated in either financial services, technology, manufacturing, retail, healthcare, public sector, or critical infrastructure (such as telecoms, energy, utilities, and transportation).

C-suite is defined as those who hold chief, global, head of department, or director roles, while reports are those who hold a manager, administrator, analyst, team lead, or officer role.



28%

UNDERSTAND HOW MACHINE LEARNING
AND DEEP LEARNING ARE RELATED

50%

OF CRITICAL INFRASTRUCTURE
ORGANIZATIONS FACED AI-POWERED
ATTACKS IN THE LAST YEAR



Top Findings



More SecOps teams rely on AI, but complexity causes confusion

- 86% of organizations are using more AI in their cybersecurity operations.
- 79% of businesses have fully or mostly implemented AI into their security tools, technologies, and workflows.
- However, fundamental AI definitions are misunderstood, with 38% unaware of or unable to define the relationship between machine learning and deep learning.



AI-powered cyber attacks drive urgent shift toward prevention-first strategies

- 46% of organizations experienced more targeted phishing attacks this year, and 43% fell victim to more deepfake impersonations.
- 83% cited attacks on local or cloud storage as a top risk to their organization, second only to phishing attacks (84%).
- 64% say the C-suite is pressing for a prevention-first strategy, and 82% of organizations have shifted strategic focus towards prevention this year.



Pace of evolution in technology and compliance increases SecOps strain

- 69% say AI and other emerging technologies are responsible for increasing SecOps burnout.
- 32% say there are too many AI-related regulations to keep up with.
- 43% are very or extremely concerned about the future impact of quantum computing on SecOps workloads.

Eye on AI: Understanding AI terminology

Artificial Intelligence (AI)

Artificial Intelligence, also known as AI, is an umbrella term covering all methods and disciplines that result in any form of intelligence exhibited by machines. Nearly all software that's currently being used in just about all industries employs some form of AI, even if it's limited to some basic, manually coded procedures.

AI Agents

An AI agent (also referred to as agentic AI, or self-driving AI) is an autonomous entity that perceives its environment through sensors and acts upon that environment through actuators to achieve its goals. These goals can range from simple tasks to complex problem-solving.

Adversarial AI

Adversarial AI refers to cyber threats that seek to inhibit the performance of AI or machine learning (ML) systems by manipulating or misleading them. It's one of many techniques (in addition to automated hacking, AI-powered phishing, evasion techniques, deepfakes, etc.) that bad actors use to inflict harm.

Dark AI

Dark AI is the programming of AI to carry out malicious activities such as cyberattacks.

Generative AI (GenAI)

GenAI refers to algorithms or deep learning models that can be used to create new content, including text, images, and video. ChatGPT, Claude, and DALL-E are examples of GenAI.

Deep Learning

Deep learning is a family of algorithms within machine learning characterized by the use of neural networks to train and analyze data. Whereas machine learning requires users to extract, compute, and vectorize a subsection of the available information to feed into the algorithm, deep learning can be trained on 100% raw data. When applied to cybersecurity, deep learning autonomously predicts threats to stop unknown malware and zero-day attacks by identifying and preventing them before they can execute.

Machine Learning

Machine learning is the leading subfield of AI, where systems are trained to recognize patterns in data or make predictions without being explicitly programmed for each task. However, the process still requires a degree of human involvement, particularly in selecting the right algorithms, preparing data, and defining which features the system should focus on.

AI, MACHINE LEARNING AND DEEP LEARNING:

WHAT THEY ARE
AND WHY THEY ARE DIFFERENT?

deep
instinct



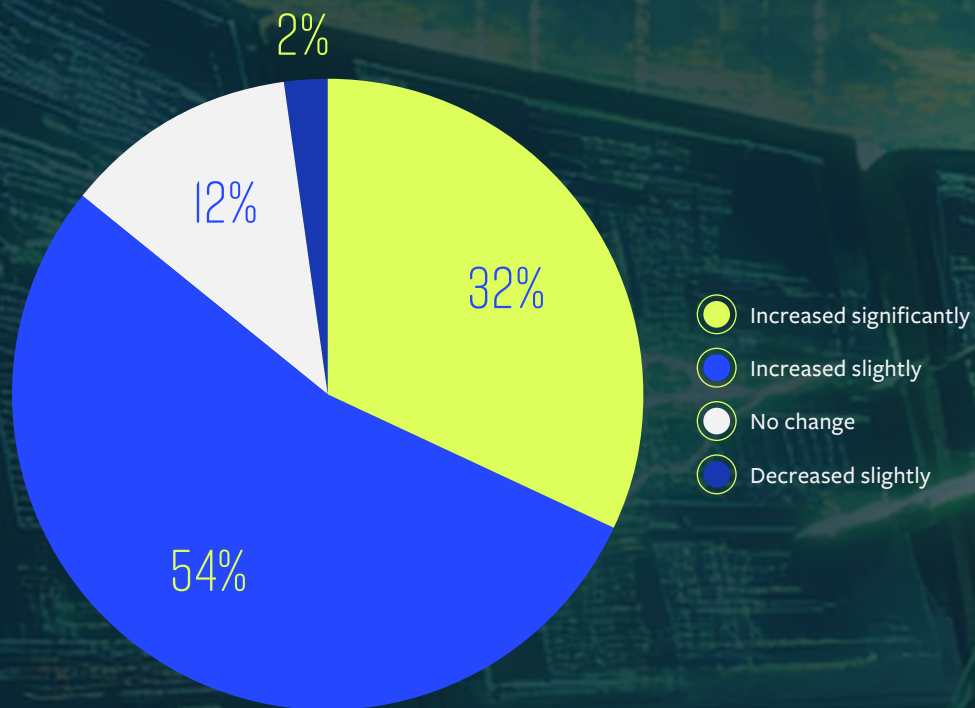
Section 1:

AI cyber adoption grows, but practical gaps remain

AI is now an integral part of most organizations' cybersecurity operations. But while the use of AI tools has become the norm – by organizations and bad actors alike – worrying gaps in understanding fundamental technologies, such as machine learning and deep learning, expose organizations to unnecessary risk.

More than 4 in 5 organizations have increased their use of AI in security operations

AI in cybersecurity has progressed from proof-of-concept to daily reality, with **86% of respondents having increased their use of AI in SecOps since last year**, and almost a third (32%) describing the increase as significant. In a threat landscape dominated by cybercriminals wielding AI with malicious intent, the only logical response is to harness its capabilities for protection.



Respondents were asked how their organization's use of AI in SecOps has changed compared to last year.

AI-powered tools are now the norm for cybersecurity

79% of businesses have fully or mostly implemented AI into their security tools, technologies, and workflows.

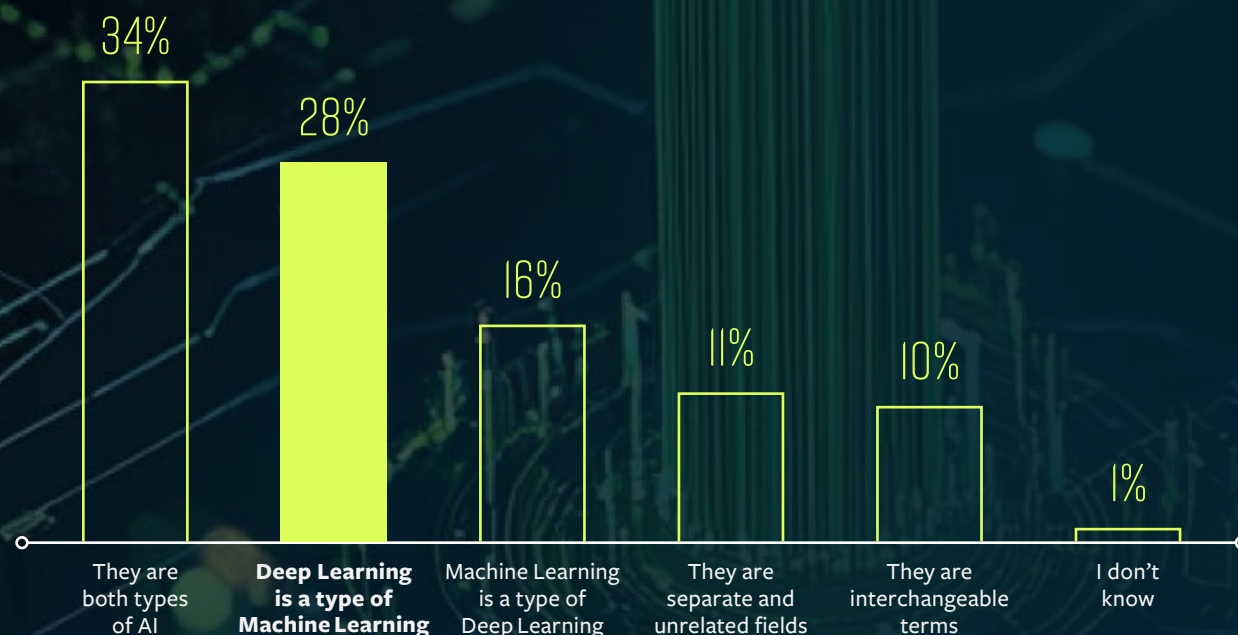


More than half of organizations (58%) use AI-powered threat detection and response tools, while another 56% use AI in identity and access management.

Respondents were asked in which areas, if any, their organization has deployed AI-powered security tools.

Two-thirds are confused about fundamental AI definitions

Combating advanced, AI-powered malware requires a strong foundational knowledge of AI, machine learning, and deep learning. However, the speed of change in this field has left IT professionals struggling to keep up: **fewer than one-third (28%) of respondents accurately described deep learning** as a type of machine learning.



38% were either unaware or unable to correctly explain the interrelation between machine learning and deep learning.

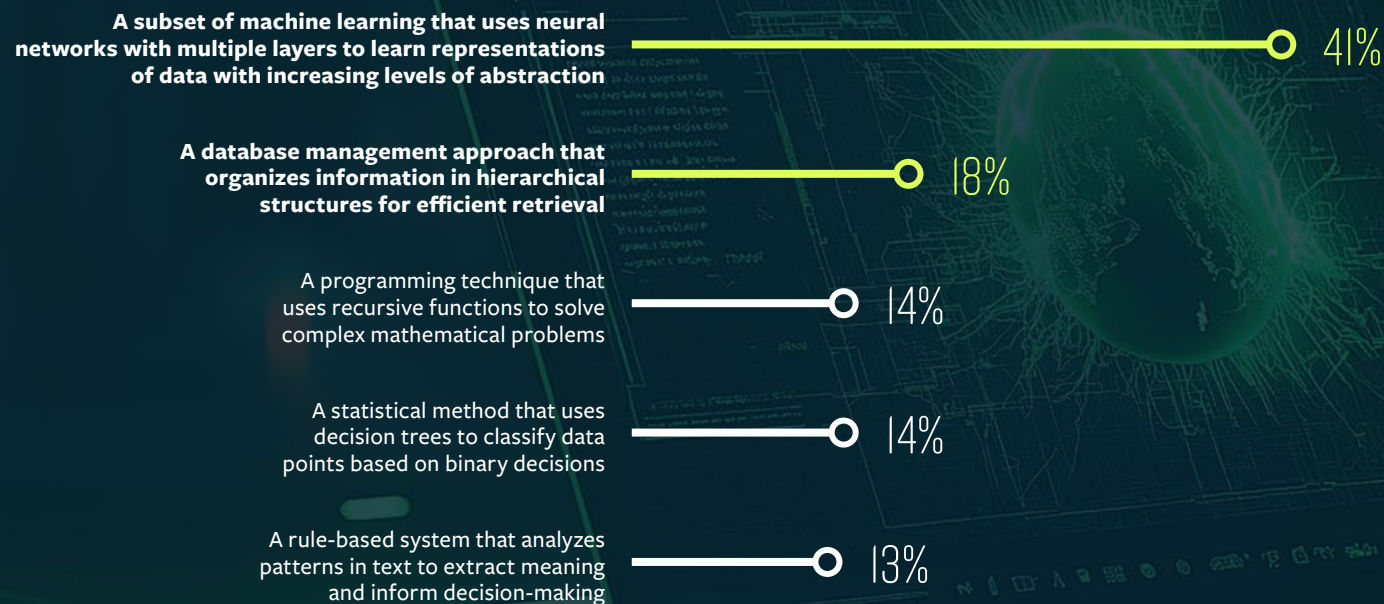
Respondents were asked which of the following best describes the relationship between machine learning and deep learning.

Another third (34%) identified both terms simply as being types of AI, without specifying further.

Those who suffered an AI-powered cyber attack last year were among those least able to accurately describe the relationship. Among the industries surveyed, organizations in the critical infrastructure sector were least likely to correctly define these key terms.

Deep learning is misunderstood by almost 60% of IT professionals

As zero-day attacks become more frequent and devastating, legacy cybersecurity tools and strategies struggle to keep pace. **Deep learning systems sit at the core of preventative strategies, but well over half (59%) of those surveyed were unable to correctly define the term.**



Respondents were asked which of the following definitions most accurately describes deep learning.

Critical infrastructure organizations again fared worst out of the sectors surveyed, with only 33% able to accurately explain what deep learning is.

Only 26% of those working in Security Operations and Analysis roles accurately described deep learning, showcasing a gap in the knowledge of one of the groups most likely to be impacted by deep learning solutions.



Focus on financial services

Financial services leads other sectors in understanding advanced AI techniques and combating AI-powered threats.

- **47% of respondents** from this industry were able to describe deep learning, compared to the average of 41% across other industries.
- **73% are increasing** their use of external security services to prepare for an increase in bad actors using AI (compared to a cross-sector average of 62%).

Two factors driving this behavior are the level of cyber threats to these organizations and compliance requirements.

- **45% suffered** an AI-powered cyber attack in the past 12 months – six percentage points above the average.
- **More than half (51%)** believe AI-related regulations are a financial penalty waiting to happen, compared to just 37% of cross-sector respondents.

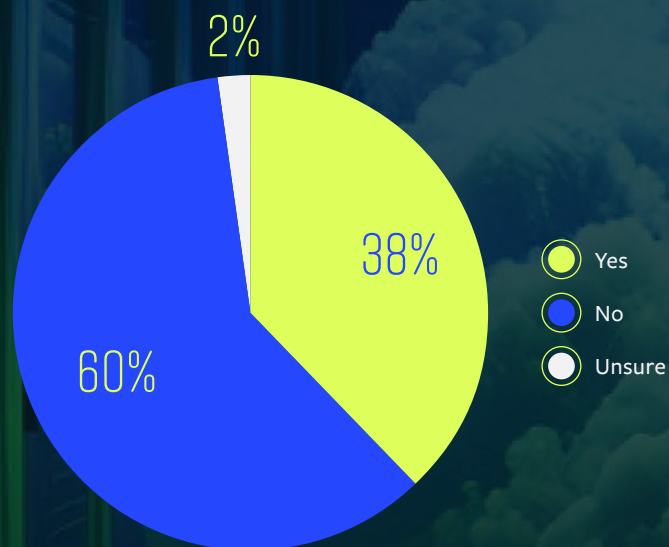
Section 2:

AI's rapid expansion increases cybersecurity concerns

Cybercriminals' use of emerging technologies is testing organizations as they work to prevent AI-powered attacks. Alongside well-recognized threats such as phishing, cloud storage is proving vulnerable to breaches, while many organizations underestimate the risk of zero-day attacks.

Cybercriminals have targeted 2 in 5 organizations with AI-powered attacks

Creation and adoption of AI tools continue to increase exponentially – in 2024, 44% of new ‘unicorn’ (privately-held, valued over US \$1 billion) companies were AI startups, compared to just 6% in 2015¹. The adoption is equally widespread among bad actors: **38% of organizations have suffered an AI-powered cyberattack in the past 12 months.**



Respondents were asked if their organization had suffered an AI-powered cyber attack in the last 12 months.

Once inside the perimeter, cybercriminals are exploiting the breach to steal sensitive or proprietary information (62%). More than half of firms attacked in this way (54%) suffer reputational damage, and around half (48%) have been affected financially.

¹ <https://www.statista.com/statistics/1557285/ai-share-of-total-unicorn-births-worldwide/>



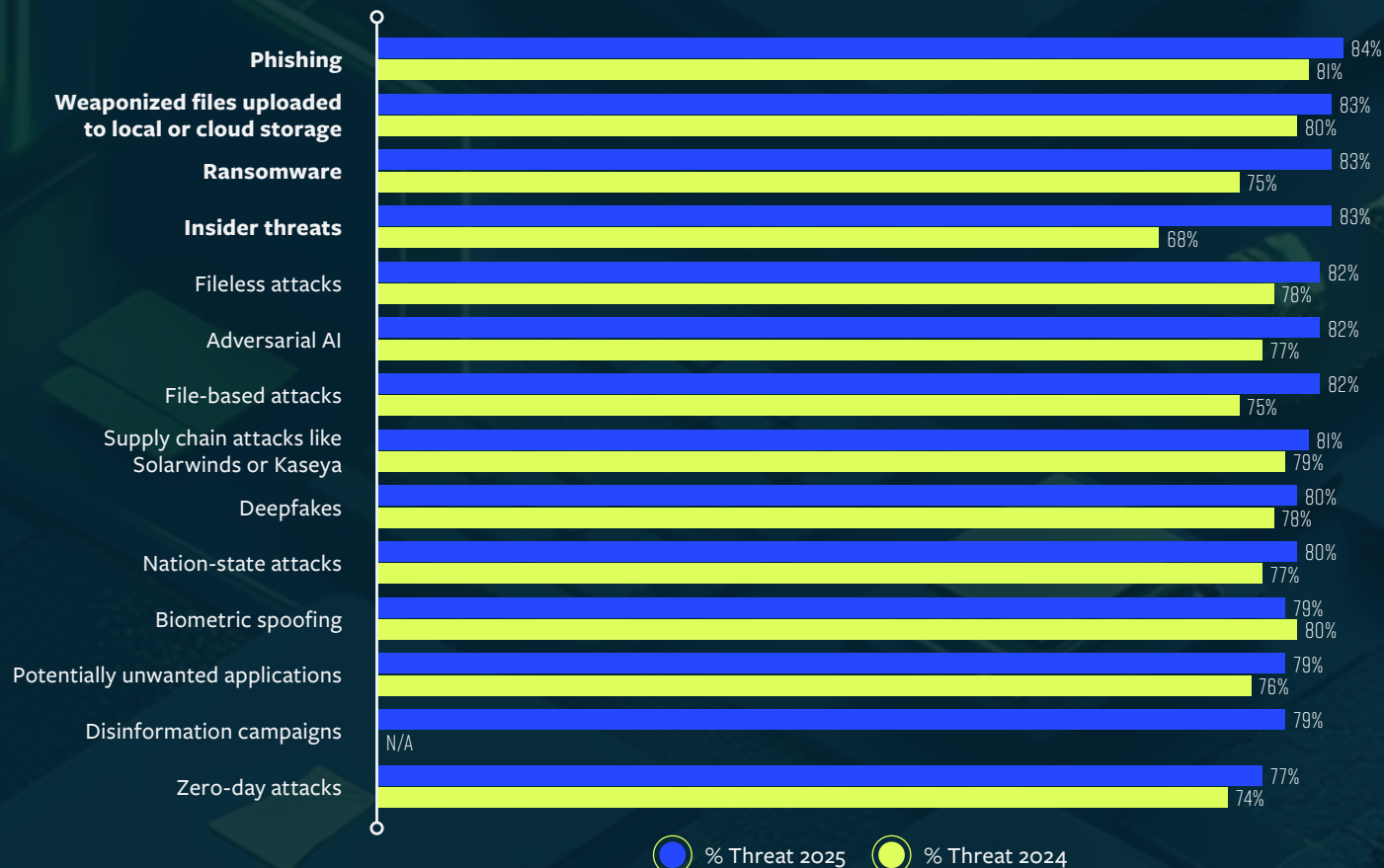
Who is most at risk from AI-powered attacks?

Bad actors are using AI to breach the cybersecurity defenses of 50% of critical infrastructure organizations – the same sector that was least likely to be able to accurately distinguish between machine learning and deep learning. Public sector and healthcare entities are least likely (29%) to be troubled by AI-powered attacks.

	%Yes
Critical infrastructure	50%
Financial services	45%
Technology & related advisories	44%
Manufacturing	32%
Retail & eCommerce	32%
Healthcare / Public sector	29%

83% say cloud storage attacks are a major cause for concern

Alongside the perennial problem of phishing attacks, **83% of organizations** say their greatest risk is less well-recognized but increasing year-on-year: **weaponized files being uploaded to file storage locations**, including local drives, Network Attached Storage (NAS), hybrid, or public cloud environments.



Zero-day attacks revealed as organizations' Achilles heel

Zero-day attacks exploit unknown vulnerabilities, bypassing reactive defenses and leaving organizations exposed. Dark AI is driving the growth in these attacks and enabling their devastating impact.

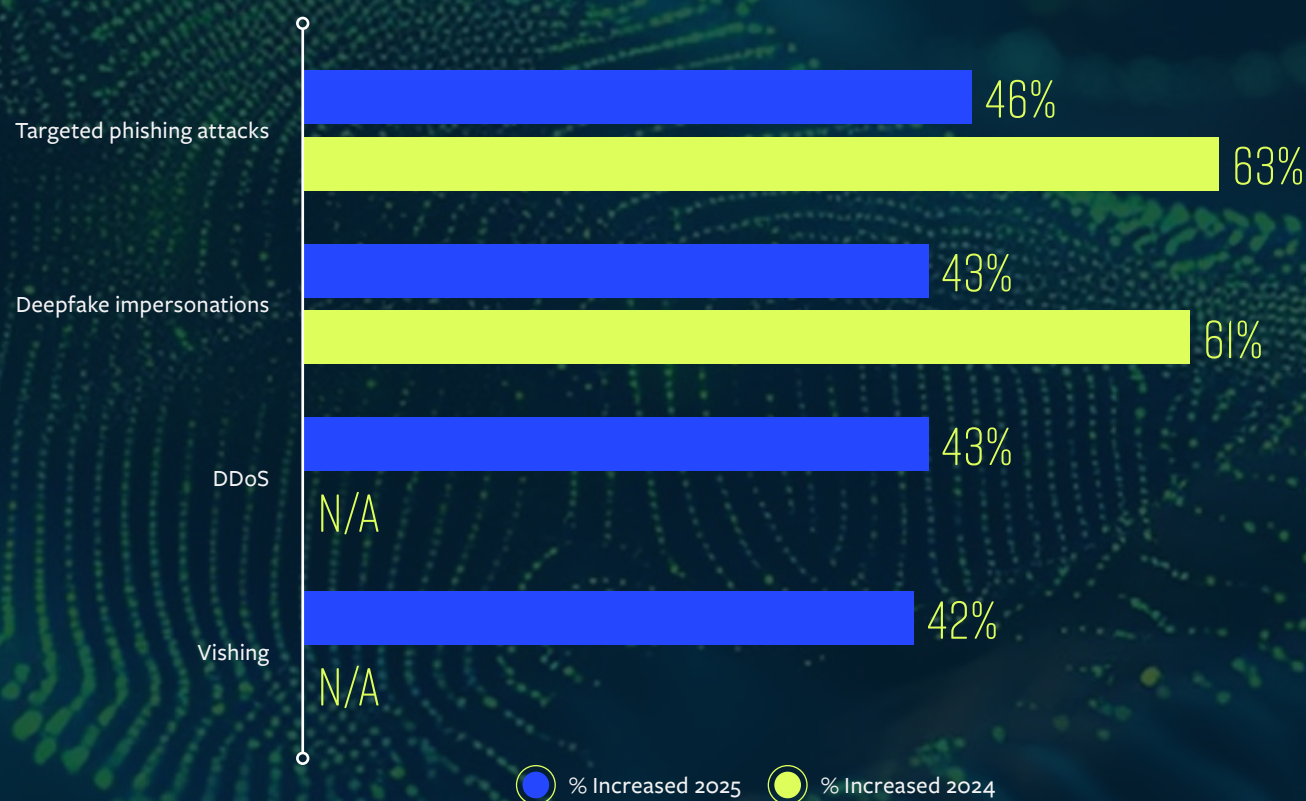
Yet the cybersecurity professionals in our survey ranked zero-day attacks as the least concerning threat. Since the potential impact is being dramatically underestimated, sufficient preemptive defenses will likely not be adopted rapidly enough, leaving many firms exposed.

Respondents were asked which of the threats listed they feel is the greatest risk to their organization's security.

Other top concerns for organizations in maintaining data security are ransomware, up 8% from last year, and insider threats, which have increased by 15% since 2024. Insider threats include both malicious activity and accidental data leaks caused by employees.

Targeted phishing and deepfake impersonations are on the rise

Almost half (46%) of respondents say their organization has suffered more targeted phishing attacks over the past year. Another 43% say they've seen a rise in the number of deepfake impersonations aimed at their business.



Who are the deepfakes impersonating?

While deepfakes featuring politicians or celebrities often hit the news, the most frequently impersonated within surveyed organizations are senior leaders within their business, such as the CEO (45%) or other C-level executives (30%).

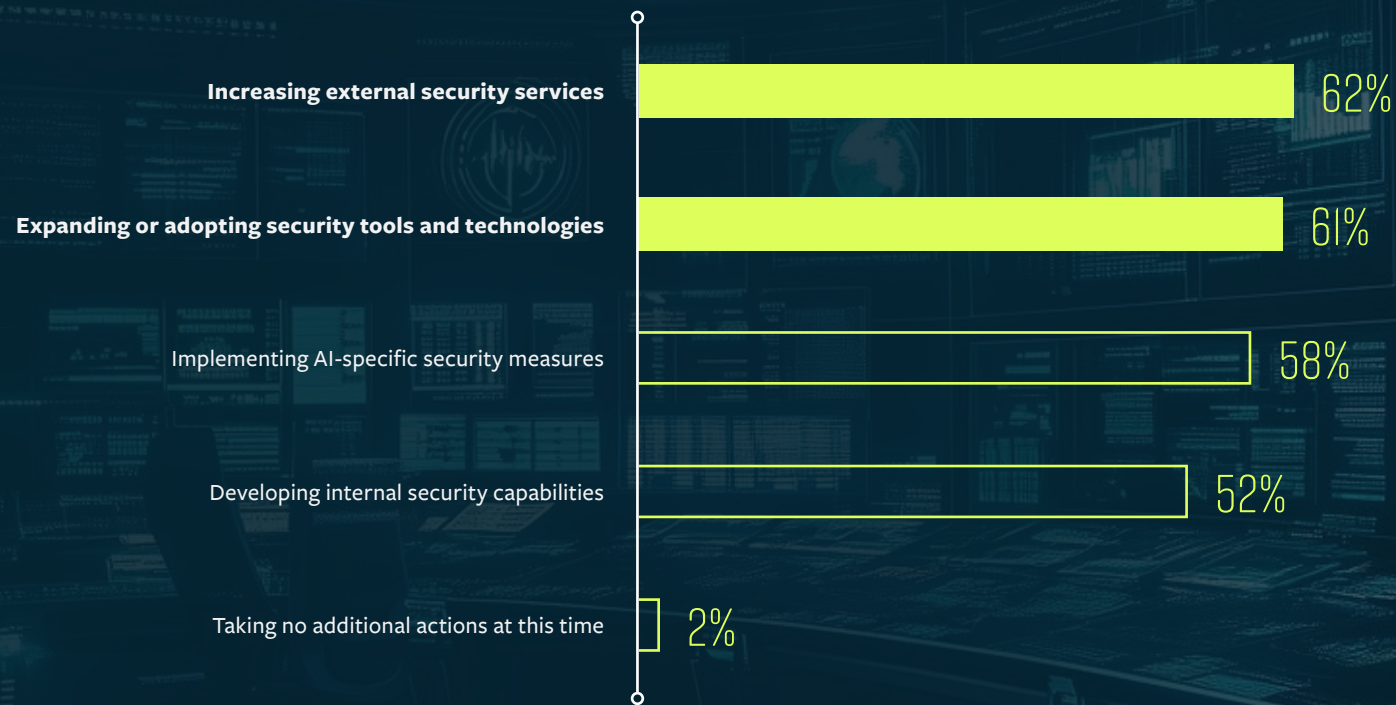
CEO	45%
Other C-level executive	30%
Contractor/third-party supplier	27%
Politician	27%
CFO	23%
Public figure and/or celebrity	15%

Respondents were asked how the listed trends evolved in terms of the frequency of incidents within their organizations over the past year.

98% take action to prepare for AI-powered threats

Activities to prepare for AI-powered cyber attacks are encouragingly widespread: 98% of organizations are assigning resources and budget to the problem.

The most popular mitigation strategy is increasing external security services (62%), followed by an expansion of existing tools or adoption of new technologies (61%).



Respondents were asked what actions, if any, their organizations are taking to prepare for an increase in AI-powered cyber attacks.

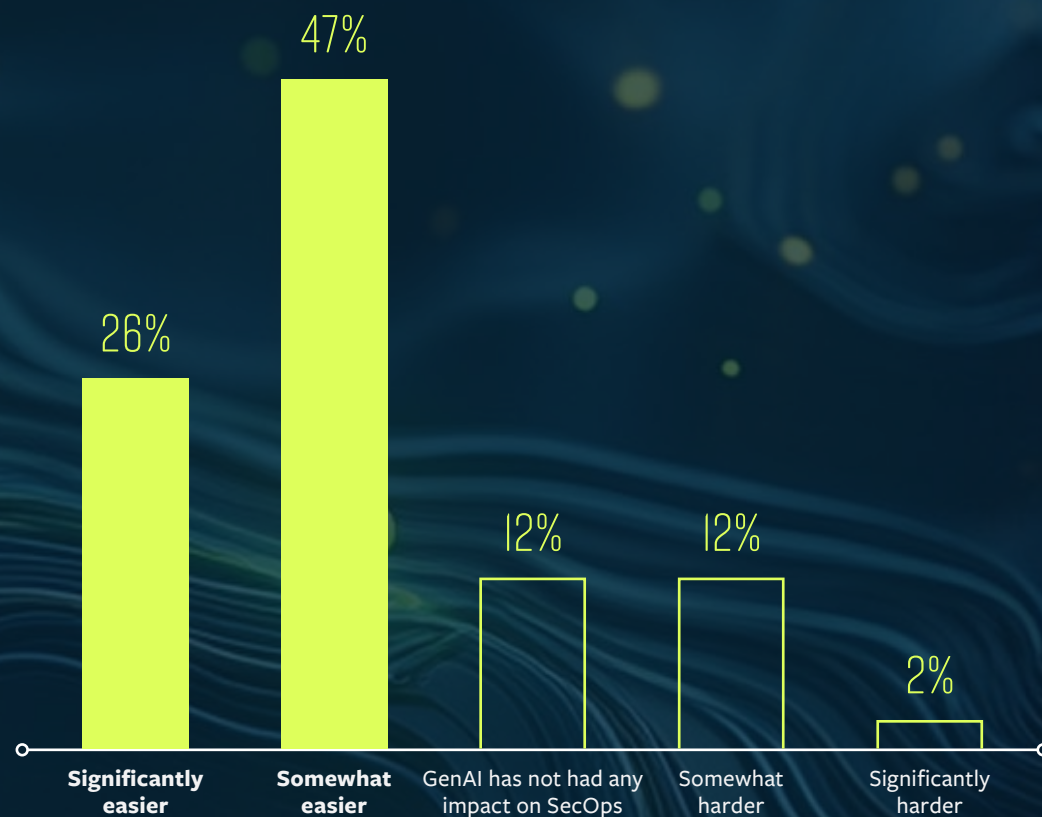
Section 3:

AI alleviates workloads, but SecOps burnout continues

AI is helping to automate threat monitoring, reduce human error, and give time back to SecOps teams. However, the pace at which emerging technologies are evolving is heaping pressure on those working in cybersecurity. A distinct shift towards preemptive measures is being pushed by the C-suite in response to the need to support internal teams and reduce exposure to risk.

Three-quarters use generative AI (GenAI) to make SecOps easier

The majority of SecOps teams (73%) feel the benefits of new technologies with more than a quarter (26%) saying GenAI makes their role significantly easier.

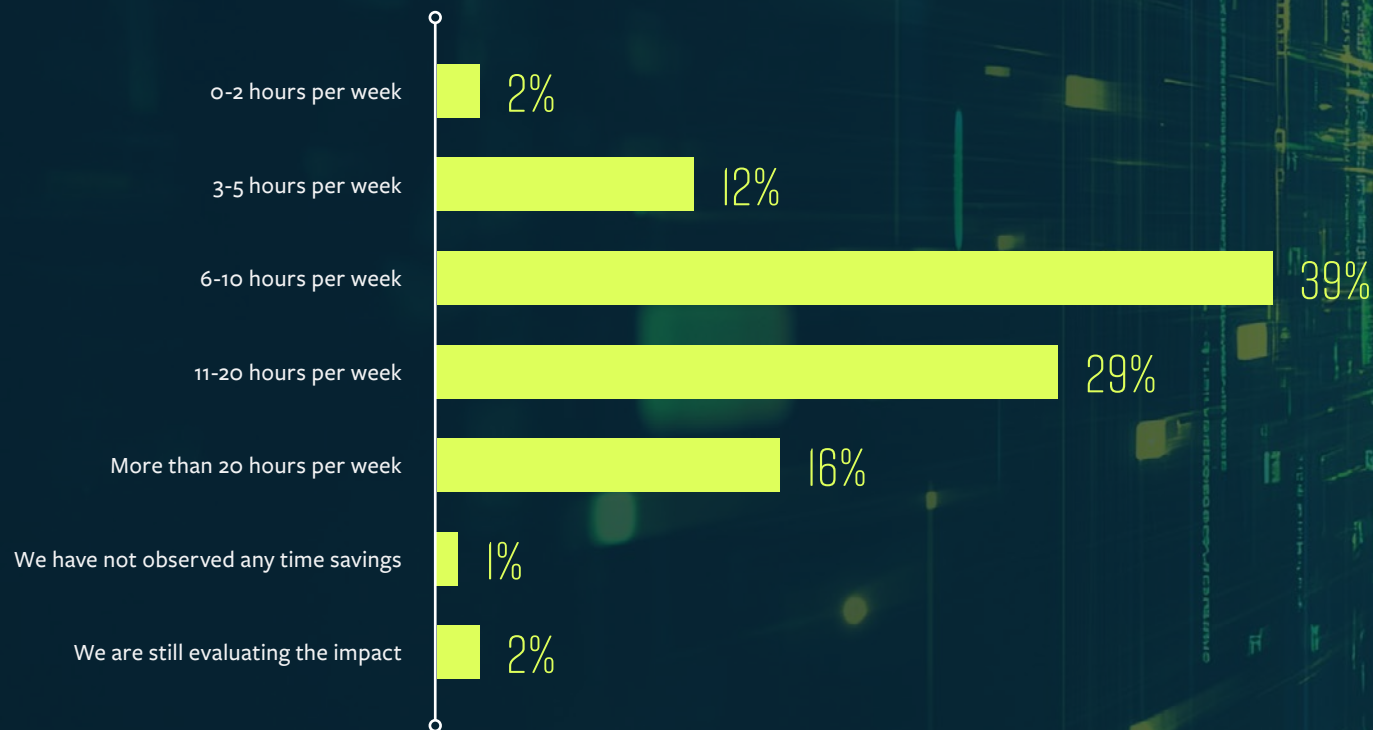


Respondents were asked to pick an answer from options ranging from “significantly harder” to “significantly easier,” to complete the sentence “GenAI is making SecOps jobs...”.

12 hours saved per week through AI-powered security tools

One notable way AI is making life easier for SecOps teams is by saving them time on manual processes.

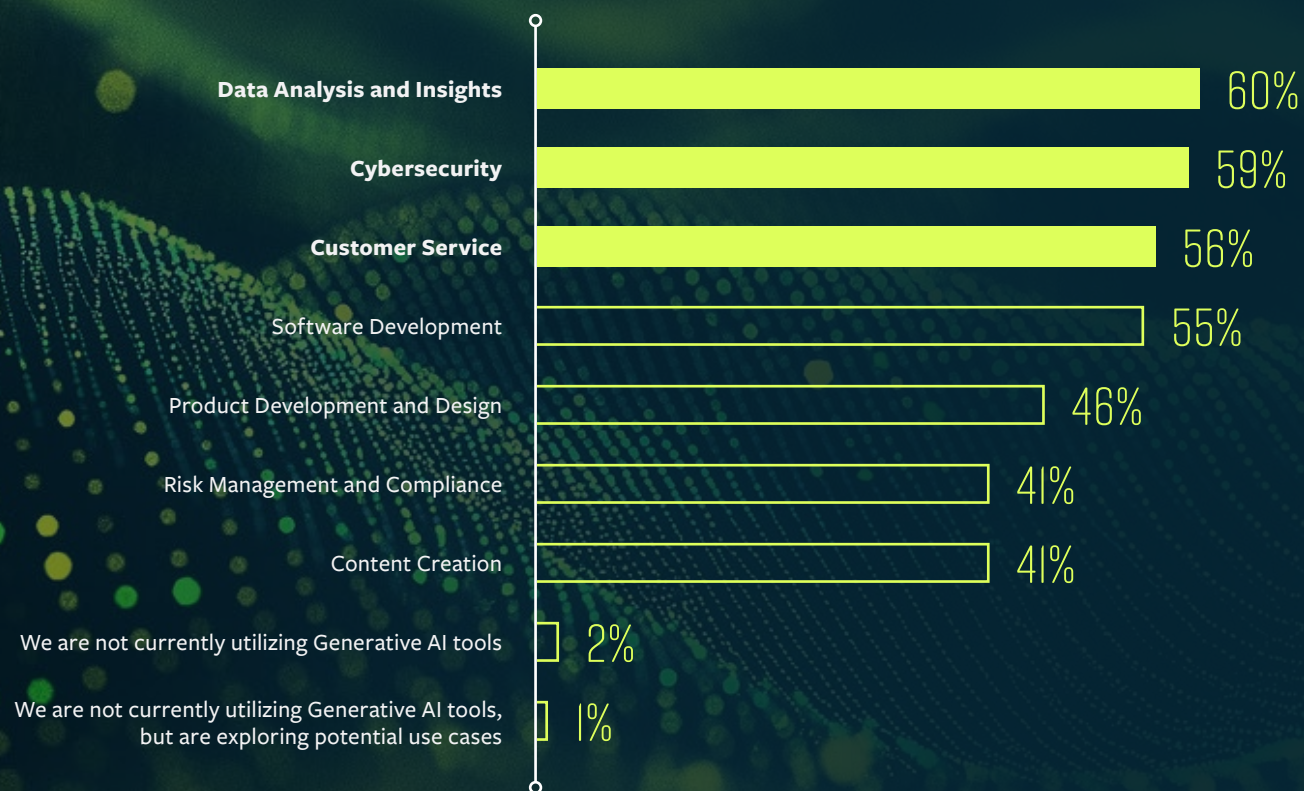
On average, organizations save 12 hours per week after introducing AI-powered security tools – rising to 15 hours for companies with over \$10 billion USD in revenue.



Respondents were asked how much time their team has saved on manual processes since deploying AI-powered security tools.

Data analysis is the top use case for GenAI

GenAI tools are being used at scale across organizations, particularly for data analysis and insights (60%), cybersecurity (59%), and customer service (56%).



Respondents were asked how, if at all, their organization is currently utilizing GenAI tools.

While GenAI tools can bring speed, efficacy, and accuracy benefits, they require oversight by SecOps teams.

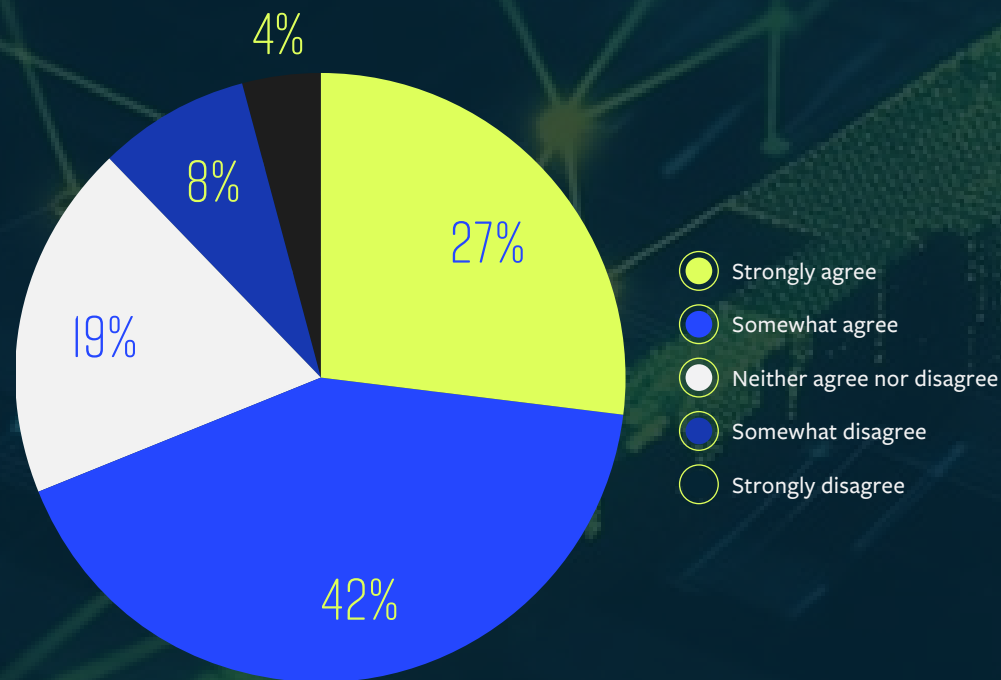
77% have a dedicated IT or SecOps team actively monitoring AI tools deployed within their organization

Who uses GenAI the most?

Those in executive leadership roles are most likely to use GenAI across all categories, particularly for cybersecurity (72%), as well as data analysis and insights (70%).

Despite the benefits, 69% say AI and other emerging technologies are contributing to burnout

GenAI may be facilitating SecOps work, but keeping up with constant cybersecurity strategy changes and new threat vectors driven by AI capabilities is a new stressor for these teams. More than 2 in 3 respondents **(69%) say AI and other emerging technologies are responsible for increasing burnout.**

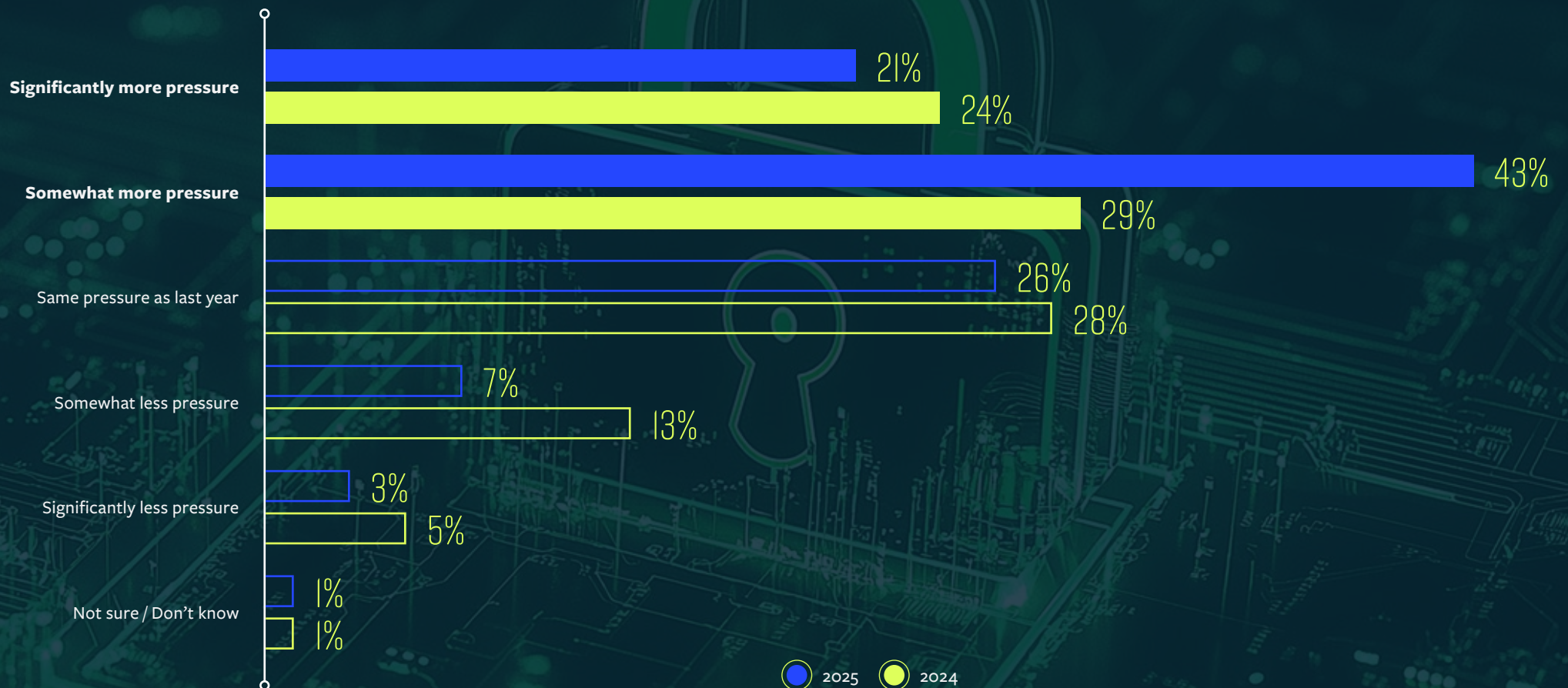


Respondents were asked to what extent they agreed with the statement: "SecOps burnout is increasing due to the rise of AI and other emerging technologies."

The C-suite calls for a shift to preemptive cybersecurity strategies

Preemptive security measures to stop cyber attacks before they happen not only reduces organizations' exposure to risk, but also alleviates the pressure on SecOps teams.

In recognition of this urgent need to switch from reacting to preventing, **nearly two-thirds (64%) say their board or C-suite is pressing them to prioritize a prevention-first strategy.** This is an increase of 11% from last year.

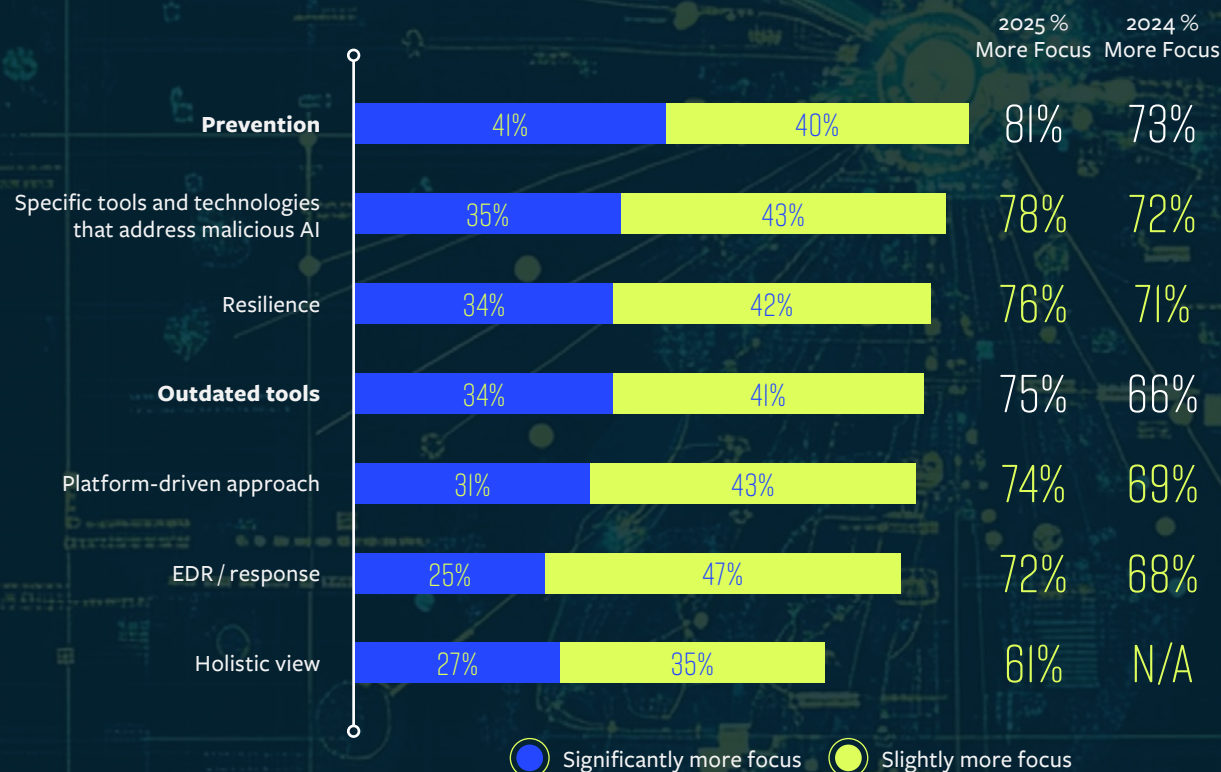


Respondents were asked, compared to last year (2024), to what extent their organization faced pressure this year (2025) from the Board and/or C-suite to prioritize implementing a prevention-first cybersecurity strategy.

Prevention tops the list of cybersecurity priority areas

Those who are revising their cybersecurity strategies put prevention top of the priority list: **81% are focusing more on preemptive security measures this year.** In 2024, only 73% said they were increasing their focus in this area.

As firms rush to adopt preventative strategies to deal with malicious AI and unknown attacks, there is also a significant rise in those looking to ditch outdated tools: three-quarters (75%) in 2025 compared to two-thirds (66%) in 2024.



Respondents were asked how their organization's cybersecurity strategy focus has shifted, if at all, over the past year (2024 - 2025) for the listed approaches.

What is Preemptive Data Security?

Gartner has identified Preemptive Cybersecurity as a new category, predicting it will be an operational requirement by the early 2030s. Preemptive Data Security is a specific implementation of Preemptive Cybersecurity, encompassing the following attributes:

Effective Unknown Threat Identification

The ability to determine the malicious nature of a file by its intrinsic behaviors and characteristics, not by pattern matching against a database of known threats.

Real-Time Zero-Day Prevention

The ability to recognize never-before-seen or encountered files as malicious and automatically quarantine or delete them pre-execution, without disrupting the flow of data through an organization.

Real-Time Insights and Explainability

The ability to scan malicious files and provide, in moments, a detailed readout of the characteristics of the planned attack, including capabilities, obfuscation methods, and any other available information.

Enterprise Speed & Scalability

The ability to seamlessly adapt to growing asset and data estates, including additional endpoints, configurations, and storage methods, without incurring speed-based bottlenecks or introducing unsustainable infrastructure growth.

Defense Across the Entire Data Estate

The ability to prevent threats wherever they are encountered in a data estate, whether at the endpoint, in applications, or in NAS or cloud storage repositories.

Purpose-Built Deep Learning Framework

Deep learning is the most advanced form of artificial intelligence. A purpose-built deep learning framework is trained on millions to billions of data points to teach it to recognize threats faster and more accurately than any other solution on the market today.

What's next for AI in cybersecurity?

While SecOps teams grapple with today's threats, tomorrow's risks loom large.



NEW THREATS ARISE

AI's rise has empowered organizations, but has supercharged Dark AI. And this is just the start. So, what's next? Threats related to quantum computing and weaponized chipsets are now appearing on the horizon, but awareness is relatively low.

43% of cybersecurity professionals have basic, minimal, or no awareness of the potential for chipsets in laptops and personal devices to be used for insider, LLM-based attacks. The same number (43%) are very or extremely concerned about the future impact of quantum computing on SecOps workloads.

Staying one step ahead requires an integrated strategy covering flexible budgets, well-equipped teams, up-to-date leaders, and a robust tech stack.



AS TECHNOLOGY EVOLVES, SO DOES THE COMPLIANCE BURDEN

While **almost half (48%) of respondents found that AI improves the accuracy of their compliance reporting**, many are nervous about the impact AI-related regulations may have on their organization.

Over a third (37%) fear AI-related regulations are a financial penalty waiting to happen, while another 32% say there are too many AI-related regulations to keep up with.



THE ANSWER IS AI – BUT NOT ALL AI IS CREATED EQUAL

The threat landscape will continue expanding, and risks will keep evolving. Reactive tools can't keep up – in this environment, firms must adopt a **preemptive data security strategy** using deep learning.

When applied to cybersecurity, deep learning autonomously predicts threats to prevent known and unknown malware, ransomware, and zero-day attacks in real time.

“Organizations must move beyond an outdated ‘detect and respond’ approach, which has proven both costly and ineffective. SecOps teams need to recognize that only with the best AI – specifically, deep learning – can they finally take the upper hand against adversaries.”

Lane Bess, CEO, Deep Instinct



Deep Instinct, the first and only preemptive data security company built on a deep learning cybersecurity framework, prevents unknown threats in <20 milliseconds, 750X faster than the fastest ransomware can encrypt. Deep Instinct Data Security X (DSX) secures data at-rest or in-motion – across cloud, NAS, applications, and endpoints.

DSX Brain, Deep Instinct's deep learning framework, prevents zero-day threats that no one else can find, with >99% accuracy and a <0.1% false positive rate. DIANNA, the DSX Companion, leverages GenAI to provide unparalleled explainability into unknown threats in <10 seconds.

For more, visit deepinstinct.com.



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